

Principles of Programming Languages

Functional Programming

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Outline

Paradigms

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Declarative Programming
Functional Programming

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Conclusion

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 - ▶ Functional Programming
 - ▶ Logic Programming

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state + modifiable variable + assignment
- ▶ *Computation proceeds by modifying values stored in locations*
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- ▶ Conventional PL are based on this computational model

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 - ▶ It is possible to compute without referring to a state. How?
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 - ▶ The computation is expressed in terms of modifications of the environment
 - ▶ “Pure” principles: do not modify the ‘state’; no side-effects

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- ▶ PLs: Haskell, Miranda (pure) + **Lisp**, ML, Scheme, ...

Some mathematical background

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- ▶ Functions = *expressible* values

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 - ▶ Application: $(\backslash x \rightarrow x * x)\ 2$
 - ▶ Higher-order functions, currying, ...

λ -calculus

- ▶ Syntax: $e ::= x \mid (e\ e) \mid \lambda x. e$
 - ▶ x is a non-terminal symbol
 - ▶ $(e\ e)$ - application
 - ▶ $\lambda x. e$ - abstraction
- ▶ Examples: 10 , $\lambda x.x$, $((\lambda x.x)\ 10)$, $(\lambda x.x+y)$

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- ▶ Examples: $Fv(\lambda x . x + y) = \{y\}$ and $Bv(\lambda x . x + y) = \{x\}$

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- ▶ Let's do it in Haskell!

Haskell example explained

► `sum = λ n. if n <= 0 then 0 else n + sum (n - 1)`

```
sum 2 → (λn. if n <= 0 then 0 else n + sum (n - 1)) 2
      → if 2 <= 0 then 0 else 2 + sum (2 - 1)
      → 2 + sum (2 - 1)
      → 2 + sum 1
      → 2 + (λn. if n <= 0 then 0 else n + sum (n - 1)) 1
      → 2 + (if 1 <= 0 then 0 else 1 + sum (1 - 1))
      → 2 + (1 + sum (1 - 1))
      → 2 + (1 + sum 0)
      → 2 + (1 + (λn. if n <= 0 then 0 else n + sum (n - 1)) 0)
      → 2 + (1 + (if 0 <= 0 then 0 else 0 + sum (0 - 1)))
      → 2 + (1 + 0)
      → 2 + 1
      → 3
```

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- ▶ But, it is possible to add it using the Y -combinator
 - ▶ Combinators are λ expressions without free variables

The Y-combinator

- ▶ In theory, λ calculus does not provide “native” support for recursion
- ▶ But, it is possible to add it using the Y -combinator
 - ▶ Combinators are λ expressions without free variables
- ▶ The Y -combinator:

$$\lambda f.(\lambda x.f (x x)) (\lambda x.f (x x))$$

Evaluation

How to evaluate $T: (\lambda x.x) ((\lambda y.y) z)$?

Evaluation

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Where is β -reduction applied first?

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Where is β -reduction applied first?

Evaluation strategies:

- ▶ Full-beta reduction
- ▶ Normal-order
- ▶ Applicative-order
- ▶ Call-by-value
- ▶ Call-by-name
- ▶ Call-by-need

Full-beta reduction

Strategy: β -reduction is applied without restriction

There are two possible β -reductions for $(\lambda x.x) ((\lambda y.y) z)$:

1. $(\lambda x.x) ((\lambda y.y) z) \rightarrow_{\beta} ((\lambda y.y) z)$
2. $(\lambda x.x) ((\lambda y.y) z) \rightarrow_{\beta} (\lambda x.x) z$

Normal-order

Strategy: the leftmost, outermost redex is reduced first

There is one possible β -reduction for $(\lambda x.x) ((\lambda y.y) z)$:

1. $(\lambda x.x) ((\lambda y.y) z) \rightarrow_{\beta} ((\lambda y.y) z)$

Applicative-order

Strategy: the leftmost, innermost redex is reduced first

There is one possible β -reductions for $(\lambda x.x) ((\lambda y.y) z)$:

1. $(\lambda x.x) ((\lambda y.y) z) \rightarrow_{\beta} (\lambda x.x) z$

Call-by-value

Strategy: the outermost redex is reduced first, but only when its right hand side has reduced to a *value*; reductions under lambda abstractions are not allowed

Value: variable or abstraction

There is one possible β -reductions for $(\lambda x.x) ((\lambda y.y) z)$:

1. $(\lambda x.x) ((\lambda y.y) z) \rightarrow_{\beta} (\lambda x.x) z$

Call-by-value

Strategy: the outermost redex is reduced first, but only when its right hand side has reduced to a *value*; reductions under lambda abstractions are not allowed

Value: variable or abstraction

There is one possible β -reductions for $(\lambda x.x) ((\lambda y.y) z)$:

1. $(\lambda x.x) ((\lambda y.y) z) \rightarrow_{\beta} (\lambda x.x) z$

Another example (the redex under the lambda abstraction is not reduced)

► $(\lambda x.x) ((\lambda y.y) (\lambda z.(\lambda u.u) z)) \rightarrow_{\beta}$
 $(\lambda x.x) (\lambda z.(\lambda u.u) z) \rightarrow_{\beta} (\lambda z.(\lambda u.u) z) \not\rightarrow_{\beta} \cdot$

Call-by-name

Strategy: the same as normal order, but reductions under lambda abstractions are not allowed

There is one possible β -reduction for $(\lambda x.x) ((\lambda y.y) z)$:

1. $(\lambda x.x) ((\lambda y.y) z) \rightarrow_{\beta} (\lambda y.y) z$

Call-by-name

Strategy: the same as normal order, but reductions under lambda abstractions are not allowed

There is one possible β -reduction for $(\lambda x.x) ((\lambda y.y) z)$:

1. $(\lambda x.x) ((\lambda y.y) z) \rightarrow_{\beta} (\lambda y.y) z$

Another example (the redex under the lambda abstraction is not reduced)

► $(\lambda x.x) \left((\lambda y.y) (\lambda z.(\lambda u.u) z) \right) \rightarrow_{\beta}$
 $(\lambda y.y) (\lambda z.(\lambda u.u) z) \rightarrow_{\beta} (\lambda z.(\lambda u.u) z) \not\rightarrow_{\beta} .$

Call-by-need

Strategy: the same as normal order, but avoid re-evaluation of terms

Call-by-need

Strategy: the same as normal order, but avoid re-evaluation of terms

Call-by-need's problem:

$$(\lambda x. x + x) ((\lambda y. y * 3) 4) \rightarrow_{\beta} ((\lambda y. y * 3) 4) + ((\lambda y. y * 3) 4) \rightarrow_{\beta} \dots$$

Call-by-need

Strategy: the same as normal order, but avoid re-evaluation of terms

Call-by-need's problem:

$$(\lambda x. x + x) ((\lambda y. y * 3) 4) \rightarrow_{\beta} ((\lambda y. y * 3) 4) + ((\lambda y. y * 3) 4) \rightarrow_{\beta} \dots$$

Solution: evaluate $((\lambda y. y * 3) 4)$ and reuse the result

Call-by-need

Strategy: the same as normal order, but avoid re-evaluation of terms

Call-by-need's problem:

$$(\lambda x. x + x) ((\lambda y. y * 3) 4) \rightarrow_{\beta} ((\lambda y. y * 3) 4) + ((\lambda y. y * 3) 4) \rightarrow_{\beta} \dots$$

Solution: evaluate $((\lambda y. y * 3) 4)$ and reuse the result

Lazy-evaluation: used by Haskell (demo)

Functional programming in Haskell

- ▶ Integers, Booleans
- ▶ Types
- ▶ Lists: head, tail, reverse, sum, infinite lists
- ▶ Lazy evaluation: examples using functions on infinite lists
- ▶ Higher-order functions: map, foldl
- ▶ Quicksort

Bibliography/Q&A

Please refer to: Chapter 11:

<https://link.springer.com/content/pdf/10.1007/978-1-84882-914-5.pdf?pdf=button> for other details and interesting stuff (e.g.,

1. local environment
2. types
3. pattern matching
4. infinite objects).

Questions?